

Car Sales Report

This project is an ai project based on Car Sales Data Set by IEEE brand ambassador
This project contains understanding, cleaning, visualization, and Exploratory Data Analysis (EDA) for the data

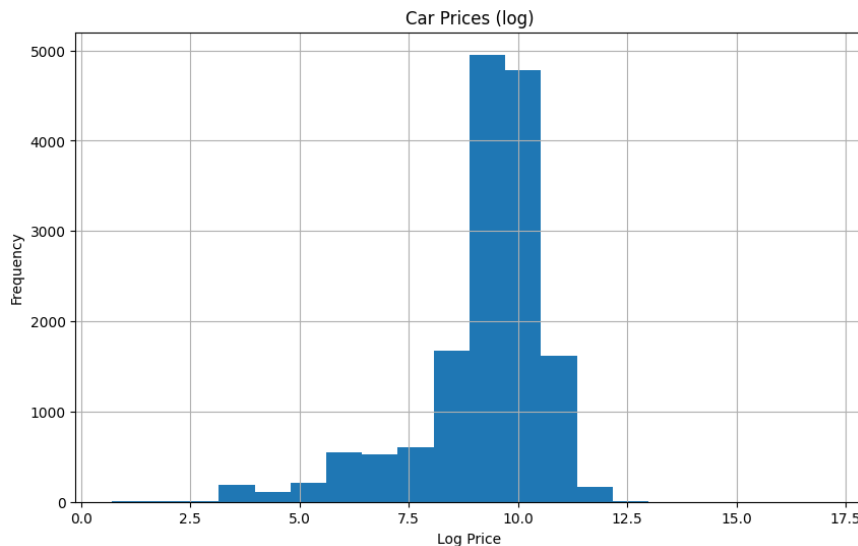
The data contain 19 features

['ID', 'Price', 'Levy', 'Manufacturer', 'Model', 'Prod. year',
'Category', 'Leather interior', 'Fuel type', 'Engine volume', 'Mileage',
'Cylinders', 'Gear box type', 'Drive wheels', 'Doors', 'Wheel', 'Color',
'Airbags', 'Random_notes']

- The data was contain null and I handle it
- Also there was conflict in data type, I solve it
- There was outlier in the data, I handle it
- And remove the duplicated rows
- There was useless features like 'ID' & 'Random notes', I removed them

Target column (price)

- There was high variance in
- the price column so I apply log to able to deal with it



missing data

- About 80% of the columns Levy, Mileage, Color were missing
- We replace the NAN with the average of each column
- All Random Notes column was missing so we dropped it

Outlier

- There were outliers in Price, Levy, Prod. Year, Engine volume, Mileage, and Cylinders contain outliers
- Some of them just because they contain a few distinct values

- And another because there was high variance in price
- So, we keep some features and the other we handle it

Some insights

- LAMBORGHINI manufacturer has the highest sum of price
- And ROLLS-ROYCE manufacturer has the lowest sum of price
- There was exposure happening in the car sales after 1990
- Black has the most car sales count then White
- Sedan is the most car soled
- Jeep category has the highest sum price
- Petrol was the fuel that has the highest sum price